

University of Economics, Prague

Faculty of Informatics and Statistics



Designing an Accessible User Interface Web Platform

BACHELOR THESIS

Studijní program: Multimedia in the Economic Practice

Autor: Ekaterina Kirichuk

Thesis supervisor: Ing. Petra Jílková, Ph.D.

Prague, May 2024

Acknowledgement

I would like to thank my thesis supervisor, Ing. Petra Jílková, Ph.D., for her diligent help and time she dedicated to me during the writing of my bachelor thesis, UX and Product Designers of Make, Marek Kříček, Daniel Munzar, Lisa Kleinman, Hedy Lemus Bird for giving their useful advice and help they have offered me, my wonderful managers Dan Etheridge and Ondřej Vocilka for their encouragement and understanding. Also I want to thank my family and my friends for mentally supporting me through that time.

Abstract

The primary goal of this bachelor's thesis is to develop an accessible low-fidelity prototype of the Make web application by integrating principles of accessible design and user-centered design, thereby strengthening digital inclusion. This thesis investigates the UX enhancement of accessibility through the development of prototypes designed to support screen readers and keyboard navigation. The theoretical framework examines legislative, ethical, and business considerations of web accessibility, alongside a review of assistive technologies and methodologies for accessibility analysis.

The practical segment of the research began with an accessibility audit of the existing Make application to identify current features and deficiencies. This audit informed the development of three innovative low-fidelity prototypes: an improved user interface (UI) that facilitates easier navigation within the application's scenario builder, a customizable accessibility settings window, and a comprehensive list of keyboard shortcuts. These features were designed based on principles discussed in the theoretical part and refined through user feedback.

Additionally, the study involved user experience testing with five participants, who provided valuable insights and suggestions for further enhancements. The prototypes not only serve as a blueprint for future high-fidelity development but also engage potential users in the iterative design process.

Overall, this thesis delivers a critical accessibility analysis of the Make application and proposes practical solutions to enhance its usability for individuals relying on assistive technologies. The outcomes underscore the importance of inclusive design and user-centered development in creating accessible digital environments.

Key words

Accessibility, prototyping, improvements, assistive technology

Abstrakt

Hlavním cílem této bakalářské práce je vyvinout přístupný low-fidelity prototyp webové aplikace Make integrací principů přístupného designu a designu zaměřeného na uživatele, čímž se posiluje digitální inkluze. Tato práce zkoumá zlepšení uživatelské zkušenosti s přístupností prostřednictvím vývoje prototypů navržených pro podporu čteček obrazovky a navigace pomocí klávesnice. Teoretický rámec zkoumá legislativní, etické a obchodní úvahy týkající se webové přístupnosti společně s přehledem asistivních technologií a metodologií analýzy přístupnosti.

Praktická část výzkumu začala auditem přístupnosti stávající aplikace Make za účelem identifikace současných funkcí a nedostatků. Tento audit informoval o vývoji tří inovativních low-fidelity prototypů: vylepšeného uživatelského rozhraní (UI), které usnadňuje navigaci v rámci nástroje pro tvorbu scénářů aplikace, okna pro nastavení přístupnosti, které lze přizpůsobit, a rozsáhlého seznamu klávesových zkratk. Tyto funkce byly navrženy na základě principů diskutovaných v teoretické části a byly upraveny na základě zpětné vazby od uživatelů.

Studie dále zahrnovala testování uživatelské zkušenosti s pěti účastníky, kteří poskytli cenné náhledy a návrhy na další vylepšení. Prototypy nejenže slouží jako modla pro budoucí vývoj high-fidelity, ale také zapojují potenciální uživatele do iterativního procesu designu.

Celkově tato práce poskytuje kritickou analýzu přístupnosti aplikace Make a navrhuje praktická řešení pro zlepšení její použitelnosti pro jednotlivce, kteří jsou závislí na asistivních technologiích. Výsledky zdůrazňují význam inkluzivního designu a vývoje zaměřeného na uživatele při vytváření přístupných digitálních prostředí.

Klíčová slova

Přístupnost, prototypování, zlepšení, asistivní technologie

Table of contents

Introduction	6
1.Design Thinking and Its Role in Creating Accessible Interfaces	8
1.1.Introduction to Design Thinking.....	8
1.1.1.Prototyping	9
1.2.Importance of Accessibility in Design.....	10
1.2.1.Accessibility in the context of Automation.....	11
1.2.2.Incorporating Accessibility to Prevent Negative Outcomes	12
1.2.3.Proactive Accessibility Measures in Design	13
2.Principles and Guidelines of Accessibility	15
2.1.POUR Principles	15
2.2.Color for Accessibility.....	16
2.3.Assistive Technologies	18
2.4.Testing Web for Accessibility	18
2.4.1.Online Tools for checking a website for Accessibility	19
3.Analysis of the Current Make Interface for Accessibility	21
3.1.Product Walkthrough	21
3.2.Product Accessibility Analysis	24
3.3.Interface Analysis Summary.....	26
4.Design and Development of an Accessible Prototype	27
4.1.Preparation to Prototyping.....	27
4.2.Designing the Accessible features	30
4.2.1.Keyboard-Accessible Scenario Control.....	30
4.2.2.Customizable Settings Window	31
4.2.3.Keyboard Shortcuts Window.....	33
4.2.4.Preparing Prototypes for Testing	34
5.Testing and Reflection	36
5.1.Testing of the low-fidelity prototypes	36
5.2.Results of the Testing.....	36
5.3.Implementing the Feedback into the design.....	38
5.4.Testing Summary	39
Conclusion.....	40
Bibliography	41
Appendix	43

Introduction

Automation platforms like Make are essential for enhancing productivity. However, accessibility has become a crucial challenge for ensuring that these tools are available to all users, including those with disabilities. This bachelor's thesis aims to enhance the accessibility of Make by introducing essential accessibility features through low-fidelity prototypes and user-centered design principles. The goal is to contribute to a more inclusive digital environment that accommodates everyone, irrespective of their physical abilities.

Make, previously known as Integromat, has gained prominence as a versatile online automation platform that allows users to create complex workflows involving multiple applications. Despite its acclaim for an intuitive, no-code interface that accommodates over 1500 applications, Make has faced criticism for its lack of adequate accessibility features. Specifically, the platform has been noted to fall short in providing necessary accommodations for users dependent on assistive technologies such as screen readers and keyboard navigation. This deficiency limits the accessibility of the platform for users with disabilities, challenging its utility and inclusivity.

The primary goal of this thesis is to develop an accessible low-fidelity prototype of the defined platform, Make, by integrating principles of accessible design and user-centered design, thereby strengthening digital inclusion. This project is conducted independently from Make's direct business activities and focuses on exploring significant accessibility issues. This initial exploration aims to inform and facilitate further enhancements of the accessibility features, potentially guiding future development efforts to make the platform more inclusive and user-friendly for all users.

The literature review for this thesis draws upon several key texts to provide a comprehensive understanding of accessibility and user-centered design, crucial for enhancing the Make platform's interface. The primary sources include Regine M. Gilbert's *Inclusive Design for a Digital World: Designing with Accessibility in Mind* (Design Thinking), which offers practical guidelines for embedding accessibility into design processes effectively. Sarah Horton, Whitney Quesenbery, and Aaron Gustafson's *A Web for Everyone: Designing Accessible User Experiences* emphasize empathy in design, crucial for understanding the challenges and needs of all users, including those with disabilities.

Jonathan Shariat and Cynthia Savard Saucier's *Tragic Design: The Impact of Bad Product Design and How to Fix It* provides insights into the negative impacts of inaccessible design and strategies for creating safer and more inclusive products. Additionally, Katie Cunningham's *Accessibility Handbook* delivers actionable techniques for developers and designers to ensure compliance with accessibility standards.

While these are the primary resources, additional materials will be utilized to enrich the research and ensure a thorough exploration of accessible design practices, supporting the thesis's goal of developing a practical, user-centered, and inclusive low-fidelity prototype.

The methodology employed in this thesis involved an initial accessibility audit of the existing Make application to identify current features and their shortcomings. Following the audit, three low-fidelity prototypes were developed: an enhanced user interface (UI), a customizable accessibility settings window, and a comprehensive list of keyboard shortcuts. These prototypes were crafted based on insights gained from the theoretical review and were refined through user feedback obtained during testing sessions with participants who use assistive technologies.

This thesis is organized into five focused chapters, each tackling a distinct aspect of enhancing accessibility for the Make platform. The first chapter, *Design Thinking and Its Role in Creating Accessible Interfaces*, introduces the concept of design thinking and its crucial role in developing accessible digital products. The second chapter, *Principles and Guidelines of Accessibility*, details the essential standards and guidelines that inform accessible design, including the Web Content Accessibility Guidelines (WCAG).

The analysis of the current Make interface is presented in the third chapter, pinpointing significant accessibility deficiencies that guide the design improvements proposed later. In the fourth chapter, *Design and Development of an Accessible Prototype*, the process of creating low-fidelity prototypes is described. The thesis concludes with the fifth chapter, *Testing and Reflection*, which evaluates the prototypes through user testing, reflects on the feedback.

This thesis significantly contributes to the field of web accessibility by providing practical solutions to enhance the usability of the Make platform. Conducted independently from Make's direct business operations, this project focuses on inclusive design and user-centered development to extend the platform's usability to a wider audience, including those with disabilities. The enhancements proposed and the low-fidelity prototypes developed offer insightful explorations into accessibility issues, serving as a foundational blueprint for future high-fidelity development within Make. These practical implementations demonstrate the profound impact of integrating accessibility features early in the design process, enhancing user satisfaction and engagement, and fostering digital inclusivity. This work not only suggests immediate improvements but also lays the groundwork for further exploration and refinement of these features in ongoing product development efforts.

1. Design Thinking and Its Role in Creating Accessible Interfaces

In the realm of digital design, ensuring accessibility is not merely a best practice but a fundamental requirement that intersects legal, ethical, and business dimensions. This chapter delves into these critical aspects of accessibility within the context of web and app development. Legally, various international statutes and national laws mandate that digital services accommodate all users, including those with disabilities, thereby setting the minimum standards for accessibility. Ethically, providing equitable access to technology reflects a commitment to inclusivity and social responsibility, recognizing the diverse needs of users and striving to eliminate barriers that prevent full digital participation. From a business perspective, accessible design opens up markets to a wider audience, enhances user satisfaction, and often leads to innovations in product development. By integrating accessibility principles into the design process, companies not only comply with legal requirements but also gain competitive advantages and contribute positively to their brand reputation. This chapter aims to outline why integrating accessibility into the design process is crucial, exploring its multifaceted impacts and the overarching benefits it brings to users and businesses alike.

1.1. Introduction to Design Thinking

Design thinking represents a dynamic and cyclic approach that teams use to deeply understand user needs, challenge preconceptions, reassess problems, and devise innovative solutions that can be rapidly prototyped and tested. This methodology is particularly effective for tackling poorly defined problems or entirely new challenges. It includes five stages: Empathize, Define, Ideate, Prototype, and Test. These stages are not always sequential; some may be conducted in parallel, out of order, or repeated as necessary (Gilbert, 2019).

The initial phase of design thinking, **Empathize**, involves comprehending the target audience and their significant preferences. This is an essential step in the design process, as it enables designers to empathize more deeply with users and streamline problem-solving. By defining a specific user group's needs, designers can create practical and user-centered solutions that meet real-world needs and increase product success. This stage usually entails conducting research, which may range from quantitative methods like Google Forms to more in-depth approaches such as user interviews (Gilbert, 2019).

In the **Define** stage, designers analyze the findings from the Empathize phase to identify the primary challenges faced by users. Here, the development of a well-crafted problem statement is crucial. An effective problem statement should be human-centered, broad enough to allow for creative solutions, yet specific enough to be manageable (Dam & Teo, 2024; Gilbert, 2019).

During the Ideate stage, designers brainstorm a wide range of ideas. This phase encourages thinking outside the box and questioning established assumptions, which can

be challenging as it requires setting aside biases and existing patterns of thought (Gilbert, 2019).

The **Prototype** stage involves creating simplified versions of the product, which allows users to test and interact with potential solutions. This step is essential as it helps identify non-viable solutions early on, saving time and resources before further development (Gilbert, 2019).

Testing, the final stage, is crucial for evaluating the usability, effectiveness, and user satisfaction of the product. Observing user interactions with the prototype helps refine the product based on real feedback. This iterative process of testing and prototyping continues until the product meets the users' needs effectively and efficiently (Conta, 2023).

Following the exploration of the design thinking process, this chapter will delve into the principles of accessibility and emphasize their importance in product development. Accessibility should be integrated from the outset of the design process, not merely added as an afterthought. By incorporating accessibility principles early, developers can create more inclusive products that cater to a broader demographic, enhancing product usability, market reach, and social inclusiveness.

1.1.1. Prototyping

In the process of designing digital products, wireframes play a crucial role in visualizing and structuring the user interface. Wireframes provide a tangible form for outlining the user experience, which is essential for effective communication and iterative refinements before the development stage. The wireframing process typically involves several stages of increasing detail, starting from low-fidelity to high-fidelity wireframes (Conta, 2023).

Low-fidelity wireframes are the initial sketches that provide a basic outline of the product's layout and user flow. These wireframes are quick to produce, facilitating rapid iteration and focusing primarily on functionality and user interaction rather than aesthetic details. This stage helps to align the team's understanding of the product's foundational structure (Gilbert, 2019).

As the design progresses, mid-fidelity wireframes build upon the earlier sketches by adding more detail to the design, refining the layout, and introducing preliminary elements of the user interface. Although these wireframes begin to look more like the final product, they still omit fine details like color and typography, allowing the design team to delve deeper into discussions about the user experience and design direction without getting bogged down by visual aesthetics (Conta, 2023).

The final stage involves high-fidelity wireframes, the most detailed and closely resembling the final product. These wireframes incorporate all visual elements, including color schemes, precise typography, and detailed interactions. This stage is critical for stakeholders and developers to get a realistic feel of the product's look and function, allowing for final adjustments before development begins (Gilbert, 2019).

The progression from low to high fidelity in wireframing ensures that each step builds upon the last, enhancing clarity and detail. This systematic approach allows for thorough testing and refinement of ideas, ensuring that by the time the product reaches development, it is well-conceived, thoroughly tested, and visually compelling. By

meticulously crafting wireframes at each stage, the design team can create a robust foundation for the final product, ensuring a smooth transition into development and ultimately a successful launch (Conta, 2023; Gilbert, 2019).

1.2. Importance of Accessibility in Design

The concept of accessibility refers to the ease with which individuals with disabilities can use or access a particular product, service, or environment. This term encompasses a wide range of adaptations and accommodations that are necessary to ensure that individuals with disabilities are not excluded from participating fully in society. Whether it is through the use of assistive technologies, the provision of alternative formats, or modifications to physical spaces, accessibility is a critical component of promoting equality and inclusion for all members of society. By making products, services, and environments more accessible, individuals with disabilities can fully participate in social, economic, and cultural activities, thereby enhancing their overall quality of life.

The term Accessibility can be used in multiple contexts; however, in this paper, we will focus on Accessibility in the web environment. It is often said that it is difficult to consider all disabilities and that it is impossible to address all of them; however, Gilbert (2019, p.114) notes that if all disabilities cannot be considered, priority should be given to blindness as it addresses approximately 80% of accessibility issues. Apart from the ethical side of the problem, the World Wide Web Consortium (W3C) has established global standards for users and developers, which touch on the legislative aspect of Accessibility. W3C defines "web accessibility" as technology designed and developed so that people with disabilities can interact with web pages and contribute to their content (Gilbert, 2019).

With the goal of providing a single shared standard for Accessibility on the web, people and organizations from all around the globe created the Web Content Accessibility Guidelines (WCAG). The WCAG documents serve as a valuable resource for enhancing web content accessibility for individuals with disabilities. The website clarifies that it is not meant to be an introductory course on accessibility but rather a comprehensive database for those seeking to implement technical standards. This includes web content developers, authoring tool creators, accessibility evaluation tool developers, and others interested in or in need of such standards.

On their webpage, w3.org, users can find at least 70 laws regarding Accessibility worldwide. The Czech Republic is also on the list with the "Act No. 99/2019 Coll., on the accessibility of websites and mobile applications and on the amendment of Act No. 365/2000 Coll., on public administration information systems and on the amendment of certain other laws" (Gilbert, 2019). The act states that the state, higher education institutions, schools, and educational facilities are responsible for ensuring their digital services are accessible for people with disabilities, making such services perceivable, operable, understandable, and robust.

According to Gilbert (2019, p.50), web accessibility benefits not only individuals with permanent disabilities but also those with temporary or situational impairments, as well as people without disabilities. Examples include struggling to see a low-contrast screen on a smartphone in bright daylight, experiencing changes in vision due to aging, or dealing with a slow internet connection or limited bandwidth. Gilbert highlights the critical

importance of ensuring accessibility in every digital product, especially given that almost 3 billion people are currently using the internet.

Accessibility is crucial not only for ethical and legal reasons but also from a business perspective. As IBM's Thomas John Watson Jr. famously stated, "*Inclusive design is good business.*" Gilbert (2019, p.128) notes that "*involving disabled individuals early in product development and marketing can open doors to a multi-billion dollar market*". Moreover, satisfied users are more likely to engage with a business, underscoring the significance of enhancing user satisfaction through accessible design (Gilbert, 2019).

Horton and Quesenbery (2015) also emphasize that implementing accessibility should begin at the earliest stages of design. By adopting an accessibility-first approach, design teams can ensure that their products are usable by the widest possible audience, thereby advancing both usability and inclusiveness from the outset (Horton & Quesenbery, 2015).

This integrated approach to accessibility not only meets diverse user needs but also aligns with best practices in digital design, fostering greater user engagement and satisfaction across a broad spectrum of the population. Previously, we explored the concepts of Design Thinking and Accessibility, investigating how these ideas intersect. In *Web for Everyone*, Horton and Quesenbery (2015) advocate for a broader view of design that goes beyond aesthetics. They introduce the concept of a "universal web," designed to be accessible for people of all abilities, attitudes, and aptitudes. This idea aligns with the principles of universal design established in 1996, which aim to create products and environments accessible to the widest range of people, minimizing the need for adaptations or specialized designs. Coupling Design Thinking with user-centered design, the Web Content Accessibility Guidelines (WCAG), and POUR principles enables designers to develop web content and applications that are inclusive to all, including those with disabilities.

WCAG + Universal Design + Design Thinking = A Web for Everyone.

1.2.1. Accessibility in the context of Automation

In today's world, automation platforms have become irreplaceable for increasing productivity and streamlining workflows across different parts of the business. By prioritizing accessibility, platforms like Make can extend to a profitable market and enable a wider group of users to benefit from their automation services.

A recent survey by WebAIM (Web Accessibility In Mind) found that 95.9% of home pages had detected WCAG 2 failures. The survey states that those are only the errors that were found automatically. The most common accessibility issue was the use of low-contrast text, with 81% of the home pages displaying text below the accessibility thresholds set by WCAG 2 AA (The WebAIM Million, 2024). This highlights a widespread issue in web accessibility, underlining the urgency for platforms like Make to lead by example in creating accessible interfaces. By doing so, Make has an opportunity to expand its user base and increase user satisfaction across a diverse spectrum of abilities, ensuring that everyone, regardless of their physical or cognitive abilities, can take full advantage of automation to optimize their workflows.

Recent research from Accenture also shows that companies leading in disability inclusion drive 1.6 times more revenue, 2.6 times more net income, and achieve twice the profit. They also mention that since 2018, with the increased opportunities to work from home, the number of people with disabilities in the workforce has risen from 29% to 37% (Accenture, 2023). As Make's main target audience is the corporate workforce, the question of inclusion becomes even more important. For automation platforms, making their tools more accessible can not only increase their user base but also contribute positively to their brand image and market value.

1.2.2. Incorporating Accessibility to Prevent Negative Outcomes

In this part of my paper, we will examine examples of how poorly designed products can significantly impact the lives of ordinary people. In their book "Tragic Design", Shariat and Cynthia Savard Saucier (2017) discuss how poorly designed products can lead to severe consequences, including fatalities. They note that accidents involving advanced technologies often result from a combination of factors, including organizational, managerial, technical, and sometimes political aspects. Regardless of how flawlessly software is engineered, it may still behave unpredictably under certain conditions (Shariat & Saucier, 2017).

While Make may not directly cause physical harm or death, it is crucial to recognize its potential to cause emotional distress to its users. By acknowledging the emotional dimensions of design, we can strive to create products that minimize harm and enhance users' quality of life.

The most common form of harm inflicted by software is emotional. People are typically more affected by negative experiences than positive ones, and a negative incident cannot simply be resolved with a quick phone call, fast email, or social media response from a brand. One straightforward approach to ensure that your design does not emotionally harm anyone is to make it polite.

Impolite technologies often exhibit characteristics such as selfishness, laziness, gluttony, and attention-seeking, which disrupt user activities by shifting focus, demanding excessive effort, and providing minimal value (Shariat & Saucier, 2017). These technologies might run persistently in the background, consuming bandwidth and system resources without the user's active engagement. This issue is not confined to lesser-known software; even products from established companies with extensive design teams can display these traits. For instance, the Xbox gaming console forces updates on users without offering the option to postpone or skip them, leading to significant inconvenience due to lengthy download and installation times.

Various assistive technologies have been developed to enhance web accessibility for people with disabilities. However, issues like incorrect language settings in speech synthesizers can lead to mispronunciations and misunderstandings. Although some screen readers support multiple languages, they often require additional language packs or voices, which users must manually install and configure (Matuzovic, 2024). Ensuring accurate language settings is essential for making web navigation more accessible to individuals who rely on these technologies.

Polite technologies prioritize user autonomy and preferences to ensure respectful interaction between the user and the software (Shariat & Saucier, 2017). They operate on principles that emphasize user experience and trust, seeking explicit permission before performing actions such as updates, information sharing, or altering default settings. This approach minimizes user frustration and builds trust.

An example of polite technology in practice is observed when setting up Google Chrome for the first time. The browser requests user consent to send crash reports to Google, respecting user choice while contributing to software improvement—a balance between user privacy and software enhancement.

Furthermore, polite technologies anticipate user needs without being intrusive and respect user decisions, minimizing repetitive requests (Shariat & Saucier, 2017). For example, e-commerce platforms may offer to adjust currency settings based on the user's location but will not repeat the offer if declined.

Polite technology also extends to the manner of communication, encouraging software to use respectful language and avoid patronizing tones. This philosophy enhances the user experience by demonstrating that even a semblance of politeness can significantly improve user satisfaction. Ultimately, polite technologies embody a design philosophy that prioritizes the respect and dignity of users, setting a standard for future software development that is both user-centered and ethically sound.

1.2.3. Proactive Accessibility Measures in Design

Shariat and Saucier (2017, p.111) underscore that accessibility should not be treated as an afterthought or a corrective measure for design flaws but should be integrated as a core principle in the design process. Echoing this sentiment, they advocate for incorporating accessibility measures right from the beginning of the design process, which proactively eliminates potential barriers and fosters a more inclusive and equitable user experience.

In order to avert poor design outcomes, several practices can be adopted. Designers should prioritize a user-centered design approach that focuses on deeply understanding and fulfilling user needs through continuous user research and usability testing. Empathy plays a critical role in this process, as it helps designers appreciate and consider the diversity of user experiences and perspectives. Further, leveraging both quantitative data and qualitative user feedback is vital for making informed design decisions that effectively address user needs.

Accessibility must be a foundational element of the design, adhered to by following established guidelines and standards from the outset. This approach not only meets regulatory requirements but also enhances user engagement and satisfaction. Additionally, adopting an iterative design process allows for ongoing refinement of the product based on user feedback, which helps in evolving the design to better meet user expectations.

The importance of good design practices should be actively promoted within organizations and the broader community to heighten awareness of the significant impacts of design

decisions. Collaboration across different disciplines is essential to ensure that the design process holistically considers technical feasibility, business objectives, and user needs.

Designers must also commit to continuous professional development by staying abreast of the latest trends, tools, and best practices in the field. Engaging with diverse perspectives and sharing acquired knowledge with the design community can lead to more robust and inclusive design solutions. Moreover, ethical considerations should be at the forefront of design decisions to avoid exploiting or harming users.

Finally, it is crucial for designers to be proactive in identifying potential negative consequences of their design choices, seeking to mitigate risks and prevent harm to users. Supporting and encouraging peers in the industry who are committed to enhancing usability and averting poor design practices is also fundamental. Through such collective efforts, the design community can contribute significantly to advancing usability and accessibility standards.

2. Principles and Guidelines of Accessibility

In this chapter, we discuss the POUR principles - Perceivable, Operable, Understandable, and Robust - for creating digital environments that are inclusive to people with disabilities. We'll explore practical strategies for enhancing accessibility in digital designs, including color modes, responsive design, and tools for testing web accessibility. We'll also look at the role of assistive technologies in creating inclusive systems. This chapter aims to equip designers with the knowledge and tools necessary to create more accessible, inclusive, and engaging digital environments.

2.1.POUR Principles

Gilbert mentions that the WCAG standards have four main success areas, known as POUR, which stand for Perceivable, Operable, Understandable, and Robust. WCAG uses these categories to measure web page accessibility (Gilbert, 2019). The first principle, **Perceivable**, states that the web resource should provide Alternative Text. This could be done in different ways, such as including alternative text so that users can easily use their screen reader technology or using captions on any audio content on a web page. Last but not least, a way to make your web more Perceivable is to include the support of Voice Recognition technology, which can be useful while entering text in a form field, as well as when navigating and activating control elements on the page (links, buttons, etc.) (Georgakas, 2023).

The **Operable** principle expects the webpage to have Keyboard Accessibility, which can be crucial for those who use only keyboards while navigating the web, users who use assistive technologies, or those with motor disabilities (Georgakas, 2023). Another operable feature is making sure that users can see the amount of time it takes them to finish a task (Gilbert, 2019).

One of the **Understandable** features is the Error Identification. The understandable principle makes sure that if there is an instance where there should be an error on the screen, it should provide the User with enough information about what he did wrong. Gilbert (2019, p.106) states that one of the best ways to handle error states is to try to plan for them, through design and testing.

Robust means that it is compatible with all types of assistive technologies, such as screen readers and screen magnification software (Georgakas, 2023).

As highlighted in previous discussions, accommodating all forms of disability in design is a complex challenge. Gilbert (2019, p.120) references a compilation of inclusive design principles developed by UK.gov, which serves as a practical framework for designers aiming to enhance accessibility across a spectrum of disabilities. These principles emphasize the integration of specific design elements tailored to the needs of diverse user groups, thereby fostering a more inclusive web environment.

For instance, when designing for users on the autistic spectrum or those with cognitive disabilities such as dyslexia, it is crucial to employ subdued color schemes, clear and concise language, and straightforward navigation. This approach helps in reducing cognitive overload and enhancing comprehension. Similarly, for individuals using screen readers or those with low vision, it is essential to ensure that digital content is structured in a logical, sequential manner, with adequate text descriptions for visual content, high contrast ratios, and legible font sizes. These adjustments make digital interfaces more navigable and readable (Georgakas, 2023).

In addressing physical or motor disabilities, the design must accommodate larger interactive elements, adequate spacing for form fields, and compatibility with assistive technologies such as keyboard navigation and voice commands (Gilbert, 2019). Additionally, for users who are deaf or hard of hearing, providing textual alternatives for audio and video content, such as subtitles and transcripts, is indispensable.

Conversely, Gilbert (2019, p.122) also outlines practices to avoid in order to prevent alienating users with specific disabilities. These include avoiding overly bright or contrasting colors for those on the autistic spectrum, steering clear of content that relies solely on visual or auditory information, and eliminating the need for precise movements in navigation which could hinder those with motor disabilities.

By embracing these recommendations, designers can significantly address approximately 80% of common accessibility issues, as most barriers faced by users with blindness also apply to other disabilities. Gilbert (2019) asserts that simple, straightforward content is key to creating an inclusive web experience that accommodates a broad audience. This approach not only aligns with ethical design practices but also enhances the functional utility of digital products, ensuring they are accessible to as many people as possible (Georgakas, 2023).

2.2. Color for Accessibility

Digital imaging, graphic design, and product design rely on different color models to represent and manipulate color information. RGB, HSL, and HSLuv offer distinct advantages and accessibility considerations. RGB is a widely used color model that combines red, green, and blue light to create a range of colors. While it is straightforward for programming and digital devices, it can be challenging for humans to intuitively manipulate due to its lack of alignment with human color perception (Boschetti & Massaron, 2018).

On the other hand, HSL is designed to be more intuitive for human understanding. This color model separates hue, saturation, and lightness, making maintaining color harmony and balance easier. It is generally easier for people to understand and adjust, aligning more closely with human vision than RGB (Boschetti & Massaron, 2018).

HSLuv is a variation of HSL that aims to achieve uniformity in color perception (Boronine, 2012). It modifies saturation and lightness to compensate for the non-linear response of the human visual system to color, making colors with the same saturation and lightness appear equally intense. As a result, HSLuv can be more accessible for tasks requiring precise color management and consistent perception across different hues, making it ideal for specialized fields that require precise color control.

In conclusion, the most accessible color model depends on the context and task. HSL offers a great balance between intuitive use and functionality for general applications and typical users. However, for specialized needs, such as professional use in technical or artistic fields where precise color control is necessary, HSLuv provides significant advantages regarding uniform color perception.

The World Wide Web Consortium (W3C) has developed the Web Content Accessibility Guidelines (WCAG) to ensure that digital content is accessible to everyone, regardless of their abilities. The guidelines classify compliance into three priority levels, namely A, AA, and AAA, each with specific requirements tailored for varying degrees of accessibility needs. Compliance with these levels is crucial for ensuring that digital content is accessible to all (Georgakas, 2023).

At Level A, the basic level of accessibility, essential actions like providing text alternatives for non-text content and captions for pre-recorded audio are required. This represents the minimum standard that all digital content should meet. Level AA addresses broader accessibility issues, ensuring that websites and apps are usable and accessible to people with disabilities. It requires a higher contrast ratio of at least 4.5:1 for normal text and includes other criteria like consistent identification of similar functionalities across web pages and captions for all live audio content (Georgakas, 2023).

Level AAA, known as the gold standard, demands the highest degree of accessibility, including a contrast ratio of at least 7:1 for normal text. AAA compliance covers more detailed and stringent guidelines, such as providing sign language interpretations for pre-recorded audio content and allowing users to postpone or suppress interruptions (Georgakas, 2023).

The contrast ratios required by the WCAG guidelines are based on the visibility needs of individuals with varying degrees of vision impairment. These ratios ensure that text is discernible against its background, enhancing readability for everyone, including those with vision impairments similar to 20/40 vision (considered moderate visual acuity loss) (Georgakas, 2023).

Achieving higher levels of WCAG compliance not only aids those with disabilities but also improves the overall user experience, making digital content more accessible in various settings, such as in bright sunlight or on devices with low brightness. The global prevalence of visual impairments, which affects an estimated 2.2 billion people, underscores the significance of designing with accessibility in mind to cater to a diverse user base (Georgakas, 2023).

To facilitate the design and evaluation of accessible content, various resources are available, including WCAG checklists, color contrast tools, and plugins for design platforms like Figma, Adobe XD, or Sketch. These tools help designers ensure their products meet the required accessibility standards before launch.

In the realm of design, choosing colors wisely is of utmost importance, especially when it comes to readability. Even the most impressive design can be compromised if the color choices make the text difficult to read. To ensure clarity, it is imperative to increase the contrast between the text and its background. However, the ideal contrast is not always

achieved by using stark black text on a white background. Sometimes, more subtle combinations can offer improved visibility and a more enjoyable reading experience (Cunningham, 2012).

2.3. Assistive Technologies

Assistive technology encompasses a variety of tools designed to help individuals with disabilities effectively perform tasks. These tools are essential for ensuring that digital content adheres to the accessibility standards set forth by the Web Content Accessibility Guidelines (WCAG). Such technologies empower individuals with disabilities to access and utilize computers and the internet.

Key among assistive technologies is the use of **screen readers**, which are software programs that help individuals who are blind or visually impaired by reading out the text displayed on a computer screen. Well-known examples include JAWS for Windows, NVDA, and VoiceOver for Mac (Georgakas, 2023).

Additionally, **screen magnification** software plays a crucial role by enabling users to enlarge text and graphics on their screens, thus improving visibility for those with visual impairments. Text readers also provide significant support for individuals with learning disabilities, eye strain, or other reading-related challenges by vocalizing text using a synthesized voice and highlighting the words as they are read. It is important to note that such software typically does not interpret non-text elements such as menus or buttons (Matuzovic, 2024).

Speech input software offers another valuable assistive technology, providing an alternative method for text entry and computer navigation through voice commands. Dragon Naturally Speaking is a prominent example of this type of software (Georgakas, 2023).

For users who cannot use a standard mouse or keyboard, **alternative input devices** are available. These include head pointers, which are mounted on the user's head and allow them to type by pressing keys with a stick or similar object, and motion tracking or eye tracking devices that control the cursor based on the movement of a target or the user's eyes. Single switch entry devices also facilitate navigation and selection on digital platforms by enabling users to activate a switch when the desired option on an on-screen keyboard or webpage is highlighted (Georgakas, 2023).

The integration of these assistive technologies significantly enhances the accessibility of digital content, making it imperative for designers and developers to consider the diverse needs of all users, including those with disabilities, in the creation of digital content. Such consideration ensures that digital environments are usable, equitable, and inclusive for everyone (Matuzovic, 2024; Georgakas, 2023).

2.4. Testing Web for Accessibility

Various straightforward evaluations can be performed to assess a web page's accessibility, focusing on different aspects that impact user experience, especially for those using

assistive technologies (White, Henry, & Arch, 2024). This can be particularly important in academic contexts, where inclusive design ensures all students and researchers can access online resources.

One of the key aspects to check is image alternative text, which ensures that all images on the page have alternative text that accurately describes their content or function. This is crucial for users who cannot see the images and need to understand their context. Another important aspect is the page title, which should be descriptive and appear in the browser's title bar or tab. This helps users identify the content and context of the page quickly, which is particularly important in academic settings where users need to locate relevant information efficiently.

Headings are also important for structuring content clearly and logically. The page should use headings that form a hierarchical outline, which assists navigation and comprehension. Additionally, color contrast should be inspected to ensure sufficient difference between text and background for readability, which is critical for users with visual impairments.

A skip link at the beginning of the page can allow users to bypass repetitive content and navigate directly to the main content. Visible keyboard focus is also important, as it ensures a visible indicator (like a border or highlight) around interactive elements as users tab through the page. This helps keyboard users know which element is active and is important for academic contexts where users must interact with content efficiently. The page should also specify its primary language in the HTML code, aiding screen readers in pronouncing text correctly. Testing the page's usability when zoomed in is also important, as all content should remain legible and functional when magnified.

For audio and visual content, captions should be confirmed on videos that sync with the audio to aid users who are deaf or hard of hearing. Text transcripts of audio and video content should be provided as an alternative information access method. Audio descriptions should also be included in videos to ensure that essential visual details are not conveyed in the main audio. Clear and descriptive labels should be provided for all form inputs that define their purpose on the page. Required fields should also be indicated for form submission.

By conducting these checks, academic institutions can assess the Accessibility of web pages and ensure they accommodate a diverse range of users, enhancing their experience and interaction with online resources (W3C, 2024).

2.4.1. Online Tools for checking a website for Accessibility

Reliable and effective tools are important when analyzing the Accessibility of online platforms. There are a number of options suitable for assessing the Accessibility of the Make.com platform. These tools offer various features that can cater to different needs and preferences.

One option is the Silktide Accessibility Checker, a free Chrome plugin that tests web pages for over 200 WCAG issues. This tool provides step-by-step guidance on how to fix detected issues and is practical for a quick and thorough check of individual pages on the selected platform.

Another option is the IBM Equal Access Accessibility Checker, which is available as a Chrome, Firefox, and Edge browser extension. This tool uses IBM's rule engine to detect accessibility issues in web applications, making it useful for developers and auditors. The tool is integrated into the browser, allowing you to identify and experiment with fixes in real time.

QualWeb is an automated accessibility checker that can assess webpage conformance to WCAG 2.1 standards. It can be used through a command-line interface or via its website, making it a flexible choice for those comfortable with command-line tools and needing a free, comprehensive accessibility assessment.

Although primarily a subscription-based tool, Axe Monitor by Deque Systems may offer some free functionalities or a trial that could be useful for initial assessments. This tool provides a deeper integration into a continuous monitoring setup and is worth exploring if this is a requirement.

Finally, AChecker is an online tool that allows you to test web pages against custom accessibility guidelines and provides detailed feedback on compliance with WCAG standards. This tool is especially useful if you want to customize or extend the accessibility checks based on specific requirements.

These tools can help identify major accessibility issues on the selected web platform, guiding improvements to make it more accessible to users with disabilities. In summary, selecting and using the appropriate accessibility checker tool is crucial to ensure that online platforms are accessible to everyone.

3. Analysis of the Current Make Interface for Accessibility

This chapter provides a comprehensive analysis of the current state of the Make interface, focusing on its accessibility features for users with disabilities. Using the Silktide Google Chrome Extension's Screen Reader Simulation, we uncover critical insights into the interface's compatibility with assistive technologies. One significant finding is that the Scenario Builder, the application's primary feature, is entirely inaccessible via keyboard, revealing a significant gap in accessibility that could hinder the usability for a substantial user base.

The interface's use of color was also evaluated, assessing alignment with the Web Content Accessibility Guidelines (WCAG). Unfortunately, findings indicate that there is non-compliance regarding color and color contrast, which could negatively impact the user experience for those with visual impairments.

Despite these issues, the primary goal is not to redesign the existing color palette directly. Instead, the development of a Low Fidelity prototype is aimed at improving the structural and navigational aspects of the Make interface. This prototype is intended to address the major accessibility barriers identified, particularly enhancing keyboard navigability and interaction within the Scenario Builder.

Through these interventions, a foundational framework is aimed to be laid by this chapter, highlighting critical accessibility deficiencies and proposing practical, low-fidelity adjustments. These interventions are sought to improve the interface's usability and ensure compliance with accessibility standards, making the tool more inclusive and effective for all users. By demonstrating the role of focused design adjustments in enhancing the overall user experience, a blueprint for future enhancements is set.

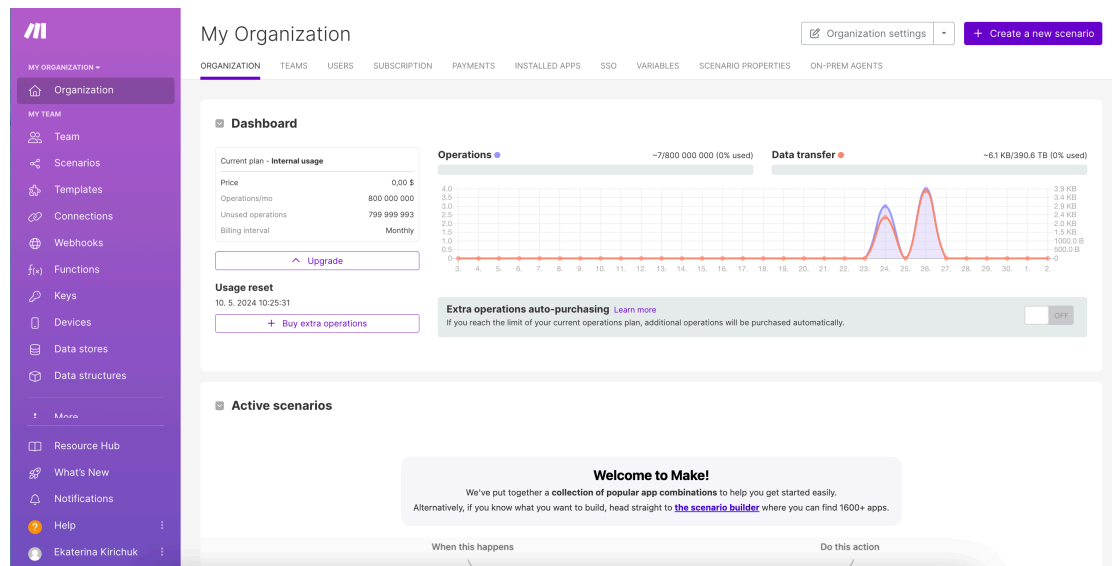
3.1. Product Walkthrough

"Make," formerly known as Integromat, is a robust online automation platform that makes it easy for users to connect apps and automate workflows. Operating in a visually intuitive environment, this platform stands out among its competitors for its ability to create complex integrations, called scenarios, that can involve multiple applications and services.

Because of its no-code nature, users can easily drag and drop elements to design automated workflows that could be useful for data transfers, receiving automated notifications, and synchronization across multiple web applications and databases. "Make" supports more than 1500 apps, including email and social media platforms and cloud storage services, that can be used in building workflows. Among its competitors, Make requires the lowest programming skills to enter and offers the most powerful solutions to users' problems.

Figure 1

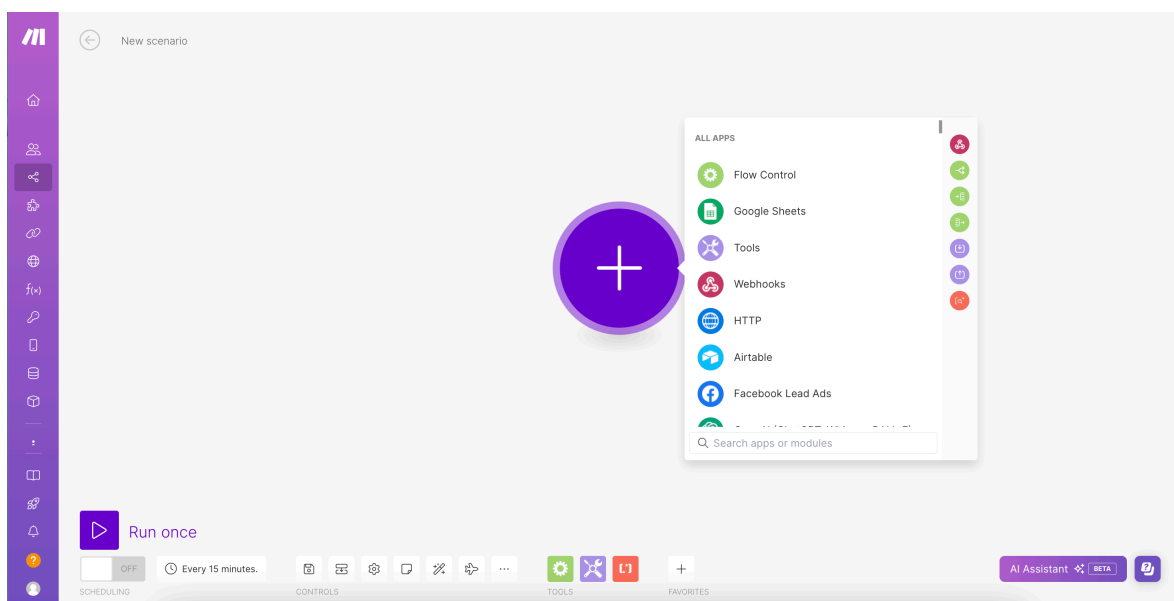
Screenshot of the Homepage in the Make app, 2024



The Make.com platform interface comprises several key components that facilitate the creation, management, and execution of complex automation workflows. The Dashboard serves as the primary entry point, providing users with an overview of their active scenarios, execution statistics, and recent history, along with access to account settings and notifications.

Figure 2

Screenshot of the Scenario Builder in the Make app, 2024

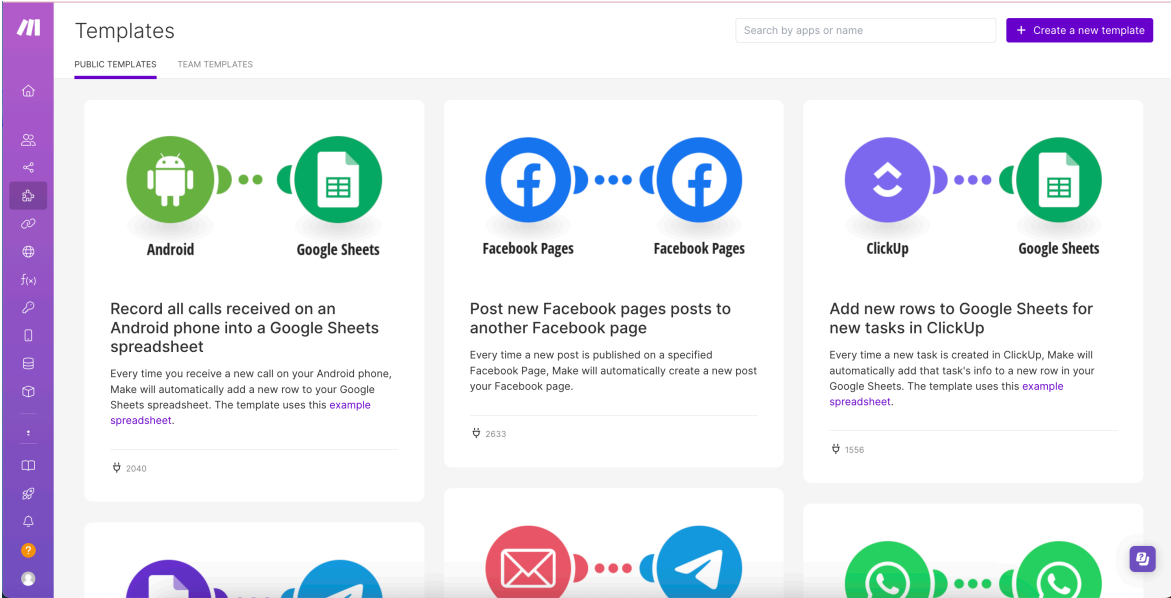


The Scenarios section enables users to create and edit automation workflows using a visual editor that consists of modules and connections. Each module within a scenario

represents an action or trigger from a specific app or service, and connections define the flow of data between these modules.

Figure 3

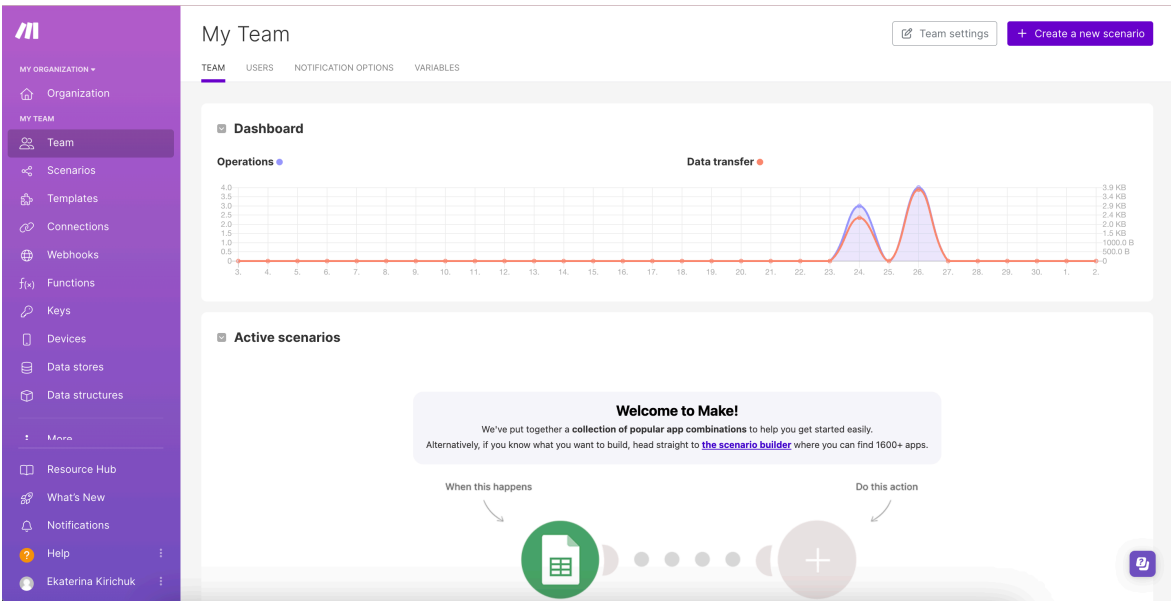
Screenshot of the Template page in the Make app, 2024



Make.com offers a library of pre-built Templates that users can leverage as a starting point for automating common tasks. The Operations section displays the number of operations performed by a scenario over a given period, with an operation being counted every time a data point is processed. The Logs section provides detailed information about the execution of each scenario, which is essential for troubleshooting and ensuring that automations run as expected.

Figure 4

Screenshot of the My Team page in the Make app, 2024



In addition, the My Team section is a crucial feature for team accounts, allowing users to manage team members, assign roles, and configure access permissions.

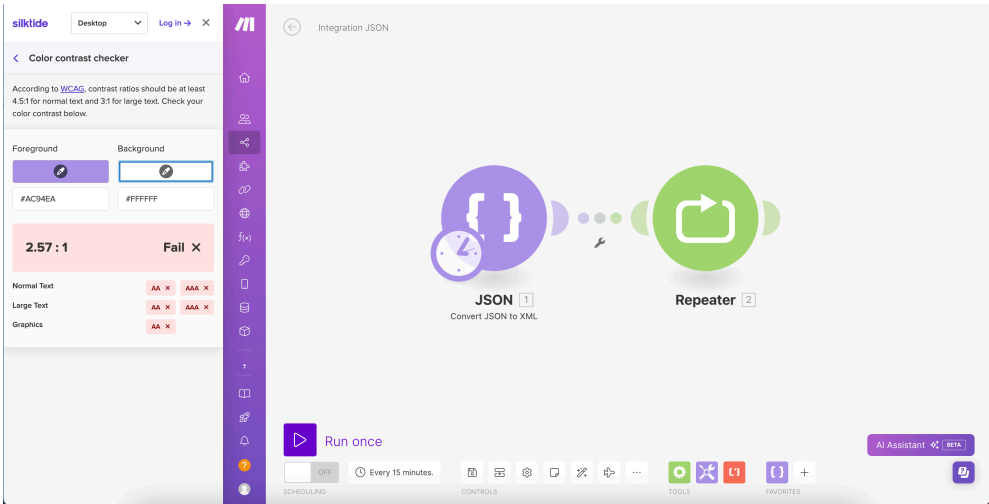
In conclusion, the Make.com platform interface is a comprehensive and intuitive tool for automating complex workflows. By leveraging its various features and functionalities, users can create and manage automation workflows that are tailored to their specific needs.

3.2. Product Accessibility Analysis

After a comprehensive evaluation of various tools, the Silktide Accessibility Checker was chosen to assess the accessibility of the Make interface. This tool stood out due to its ease of use and robust functionality, notably its capability to test web pages against over 200 WCAG issues. As a free Chrome plugin, the Silktide Accessibility Checker is an invaluable resource for identifying and addressing accessibility barriers within web environments. Its high reputation is bolstered by its recognition on official WCAG pages and alignment with accessibility standards.

Figure 5

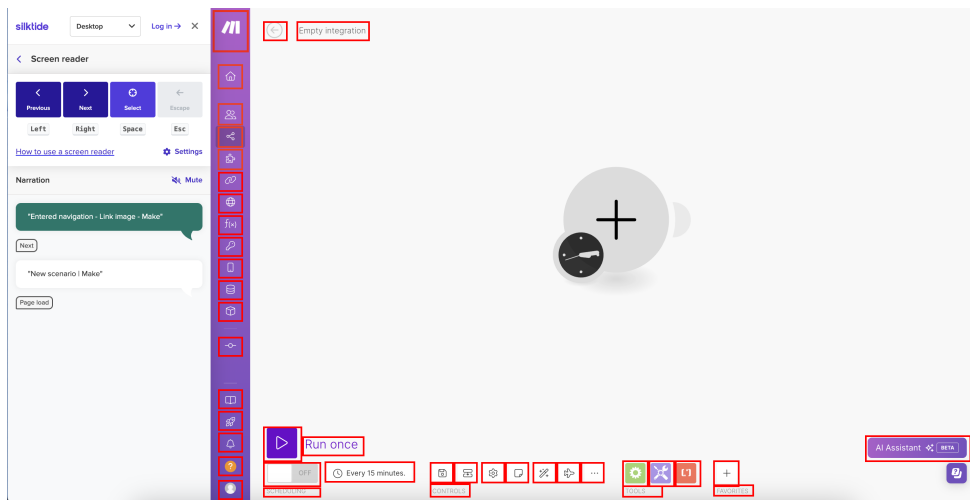
Screenshot of the Color Contrast Analysis of Make app, 2024



During the analysis, it was observed that the application consistently struggles with color contrasts across its pages, failing to meet the WCAG standards for color contrast. This issue presents a significant barrier to readability and visual accessibility.

Figure 6

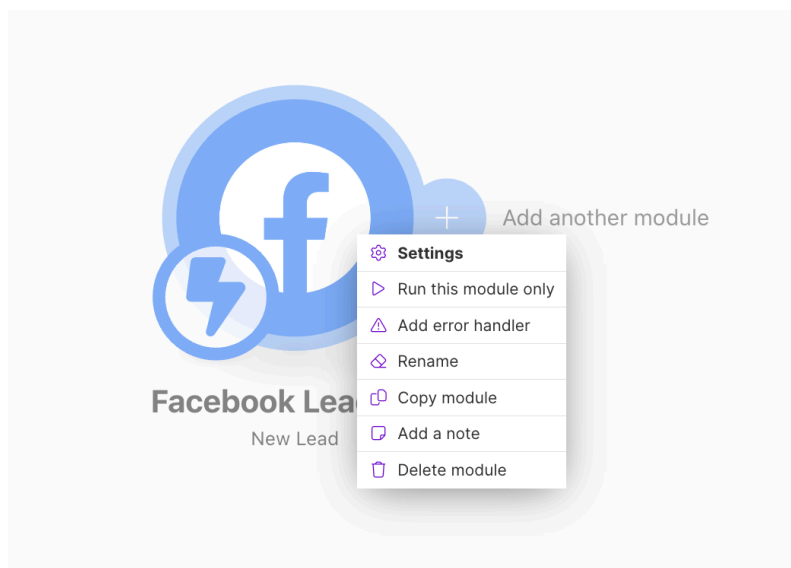
Screenshot of the Keyboard Accessibility of Make app, 2024



A more critical issue was uncovered within the app's core feature, the Scenario Builder. This visually appealing drag-and-drop interface was found to be completely inaccessible via keyboard, in stark contrast to other pages of the application that support keyboard navigation adequately. Areas of the app that are accessible with only the keyboard have been highlighted. The absence of keyboard accessibility in the Scenario Builder—a crucial component of the app—represents a significant oversight and a key area for improvement to enhance inclusivity and accessibility.

Figure 7

Screenshot of the Module Settings, 2024



Further analysis of keyboard accessibility in Make's Scenario Builder has revealed profound limitations that affect user interaction. Key functionalities such as accessing module settings, running a single module, adding new modules, renaming, and cloning

modules have been identified as inaccessible via keyboard, greatly hindering the ability to build or modify scenarios independently. Additionally, the tasks of adding notes to modules, moving modules across the board, and effectively utilizing zoom functions are all compromised without mouse interaction. These activities, essential for managing workflow scenarios effectively, pose a challenge for those reliant solely on keyboard navigation. Moreover, undoing or redoing actions, which are critical for managing changes efficiently, are found to be unachievable with a keyboard.

These significant accessibility barriers within the Scenario Builder underscore crucial deficiencies. For a tool designed to boost productivity and streamline operations, these limitations could deter a substantial portion of potential users from effectively utilizing the application. Therefore, addressing these issues is seen as imperative not only for compliance with accessibility standards but also for ensuring that all users can fully engage with the application's capabilities and benefit from its intended productivity enhancements.

Additionally, given that Make is a highly visual platform filled with animations, which serves as its main value proposition, suggestions have been made that improving accessibility could include functions to tone down colors or ideally change the theme of the app. Providing options to turn off additional animations or the transparency of some objects would also make the platform more accessible and comfortable for users with visual sensitivities or those who prefer a simpler visual experience. These enhancements would ensure that Make's visual strengths do not become barriers to usability, further aligning the product with inclusive design principles.

3.3. Interface Analysis Summary

In conclusion, this analysis has illuminated several significant accessibility issues within the Make interface, particularly within the Scenario Builder, which lacks keyboard accessibility. This finding is critical as it highlights a primary area where the application falls short in enabling all users to effectively engage with its core functionality. While the color contrast issues throughout the app also pose a barrier to accessibility, the immediate priority for enhancement lies within improving the Scenario Builder's accessibility features.

In the practical component of this project, the focus will be on addressing these key functional shortcomings by developing solutions to integrate keyboard accessibility into the Scenario Builder. By prioritizing these enhancements, the aim is to significantly improve the usability of the platform for a broader audience, including those who rely solely on keyboard navigation. Although the issues of color contrast are not the primary focus of the immediate modifications, they remain an important aspect for future updates to ensure comprehensive accessibility. This targeted approach allows for a concentrated effort on making substantial and impactful changes where they are most needed, setting the stage for further inclusive practices in subsequent updates.

4. Design and Development of an Accessible Prototype

In this section of the thesis, the design process for a low-fidelity prototype incorporating new features aimed at enhancing the accessibility of the Make interface is detailed. These improvements focus on key areas to significantly boost accessibility for users:

Firstly, modifications to the Scenario Builder include keyboard-accessible options for resizing, fitting the entire scenario to the screen, and enabling a full-screen view. This enhancement greatly improves navigation and usability for users who rely solely on keyboard input. Additionally, a new settings window is introduced, allowing users to customize the application's appearance. This customization includes options for adjusting animation and transparency levels, specifically catering to users with visual impairments. Another significant enhancement involves the implementation of a comprehensive list of keyboard shortcuts. This feature enables users who solely use a keyboard to effectively navigate and operate the application, ensuring that all functionalities are accessible without the need for mouse interaction.

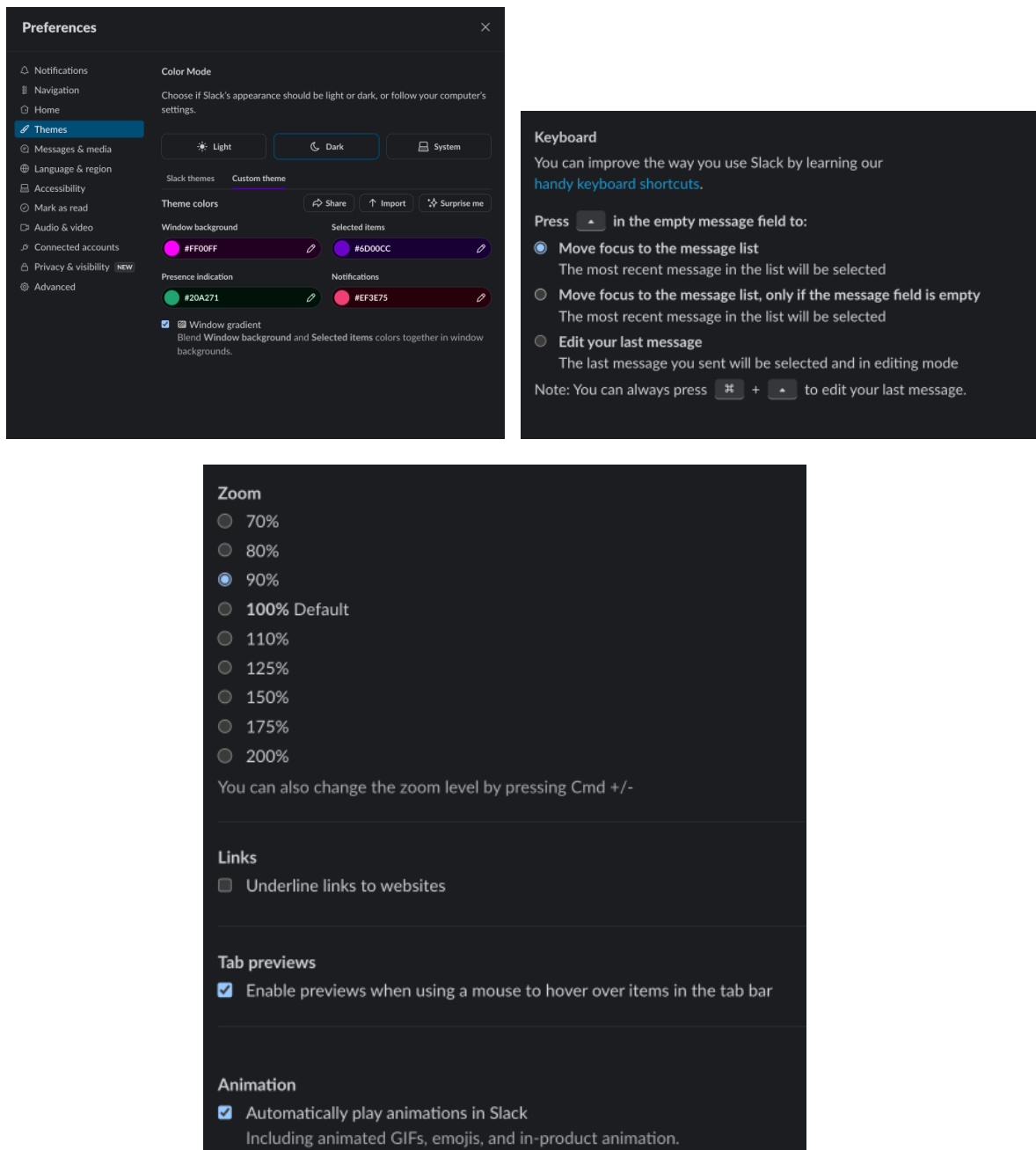
Once these features are designed, the prototype undergoes testing via five user sessions. The insights gained from this feedback are crucial in identifying necessary refinements and additional enhancements. It is important to note that the development of these features was conducted as part of the author's academic work, independent of regular business activities. Contributions and innovations detailed in this document are solely the author's initiative and part of this thesis project.

4.1. Preparation to Prototyping

In preparation for the prototyping phase of the Make interface, time was taken to gather inspiration from applications known for their exceptional accessibility features. This research involved an analysis of how various platforms, including Slack, Miro, and Adobe Photoshop, approach accessibility. Valuable insights into best practices and innovative solutions were provided through this analysis.

Figure 8

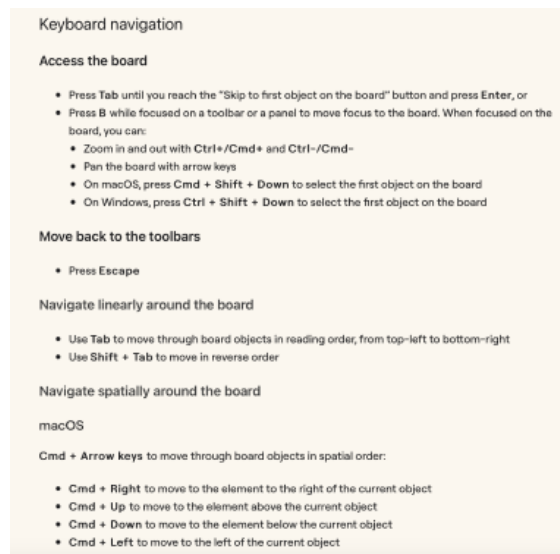
Accessibility Features of the Slack app, 2024



In Slack, enhancing accessibility is a seamless process, with users being able to customize the theme, adjust the zoom level of the interface, and access a comprehensive list of keyboard shortcuts through the preferences tab. This setup allows users to tailor their experience within the application to their specific needs directly within the user interface.

Figure 9

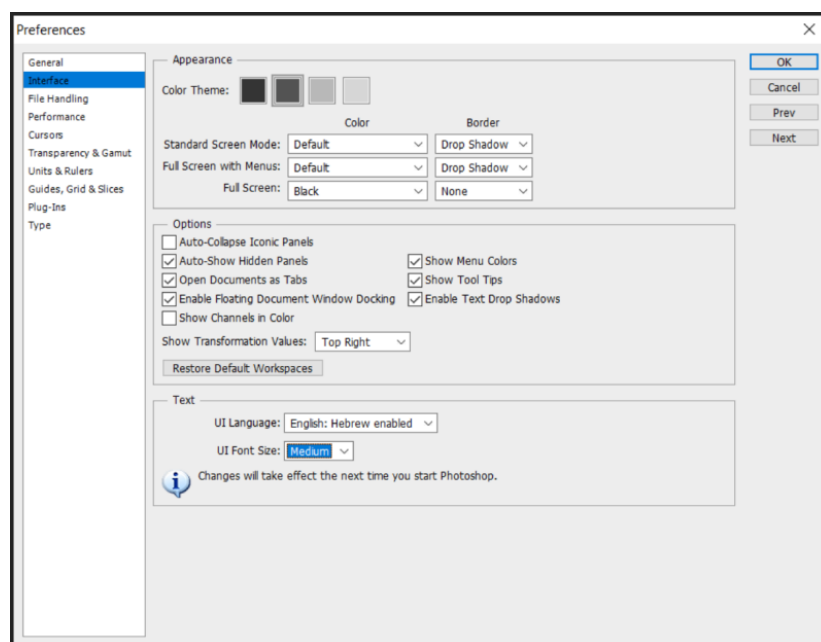
Keyboard Shortcuts of the Miro app, 2024



Miro, on the other hand, has a different approach, where users need to navigate away from the application to find information on using keyboard shortcuts. This setup could potentially disrupt users' workflow, requiring them to leave the application to seek necessary information.

Figure 10

Accessibility menu of the Photoshop app, 2024



Adobe Photoshop, in contrast, offers an integrated method for adjusting accessibility settings directly within the software. Users can open the application, navigate to the 'Edit' menu, select 'Preferences', and then choose 'Accessibility' to enable features like zoom

with scroll wheel, adjust UI font size, and activate full keyboard access. This method allows for immediate adjustments without needing to exit the application environment.

Drawing from these accessibility features, a commitment has been made to integrate similar functionalities into the Make interface. To enhance usability for users who rely on keyboard navigation, keyboard-accessible options for resizing, fitting the entire scenario to the screen, and enabling full-screen view in Scenario Builder are planned to be added. Additionally, a settings window that allows users to customize the application's appearance will be introduced, and to ensure that all functionalities are easily accessible without mouse interaction, a comprehensive list of keyboard shortcuts will be implemented.

4.2. Designing the Accessible features

After careful evaluation, Figma has been chosen for designing the low-fidelity prototypes of the Make interface. This decision leverages Figma's robust platform capabilities, which support swift design iterations and enable a seamless transition into interactive prototypes. This approach will facilitate immediate testing and refinement of the design concepts, ensuring that the prototypes are both functional and user-friendly from the outset. The following key features will be integrated:

1. Keyboard-Accessible Scenario Control.
2. Customizable Settings Window.
3. Comprehensive Keyboard Shortcuts.

These enhancements are designed to significantly improve the accessibility of the Make interface, making it more inclusive and easier to use for all users. By implementing these features, the aim is to create a more adaptable and user-centered design that responds effectively to the diverse needs of the user base.

4.2.1. Keyboard-Accessible Scenario Control

In developing the low-fidelity prototype for the first feature in Figma, a primary focus was placed on enhancing accessibility within the existing interface of the Make application. Careful consideration was given to how new keyboard-accessible controls could be naturally integrated without disrupting the flow of the current layout. By analyzing the existing user interface and features, it was determined that placing the new controls alongside existing elements would provide a seamless experience.

To execute this in Figma, a straightforward, low-fidelity representation was designed. The process began with drawing a basic rectangle to house the new controls, emphasizing simplicity to keep the prototype focused on functionality rather than aesthetic details. Inside this rectangle, intuitive icons were included—plus and minus for resizing the interface, and symbols for 'fit to screen' and 'full screen' adjustments. Additionally, a panel specifically for undo and redo actions was integrated, applying the same design principles to ensure consistency and accessibility. To reinforce accessibility and ensure clarity for all users, particularly those relying on keyboard navigation, descriptive low-fidelity text

popups were added next to each icon and the new undo/redo panel. These popups explicitly label the icons' functions, making the intended actions clear and easy to understand.

Figure 11

Low-fidelity Keyboard-Accessible Scenario Control prototype, 2024



This approach not only facilitated the integration of accessible features into the prototype but also maintained the low-fidelity nature of the design, essential for focusing on functional improvements and rapid iteration. By prioritizing accessibility in this early stage of design, the prototype lays a strong foundation for further development and testing, ensuring that the final product is inclusive and user-friendly.

4.2.2. Customizable Settings Window

In the development of the second feature for the Make interface, the groundwork for enhanced accessibility options within the application was focused on, with the structure of the windows being designed solely in low fidelity. An "Accessibility" icon and accompanying text were added to the left navigation menu, establishing a clear access point for users looking to adjust the application's settings.

Upon selecting the Accessibility menu item, a pop-up window concept was designed. This window is divided into three sections, each dedicated to different customization aspects: color mode, visual effects, and keyboard shortcuts. The layout for each section was created using simple shapes and labels to maintain a low-fidelity aesthetic and focus on structural design without interactive elements.

In the Color Mode Section, three placeholder rectangles were included to represent options allowing the user to change the interface theme to dark, light, or match the system

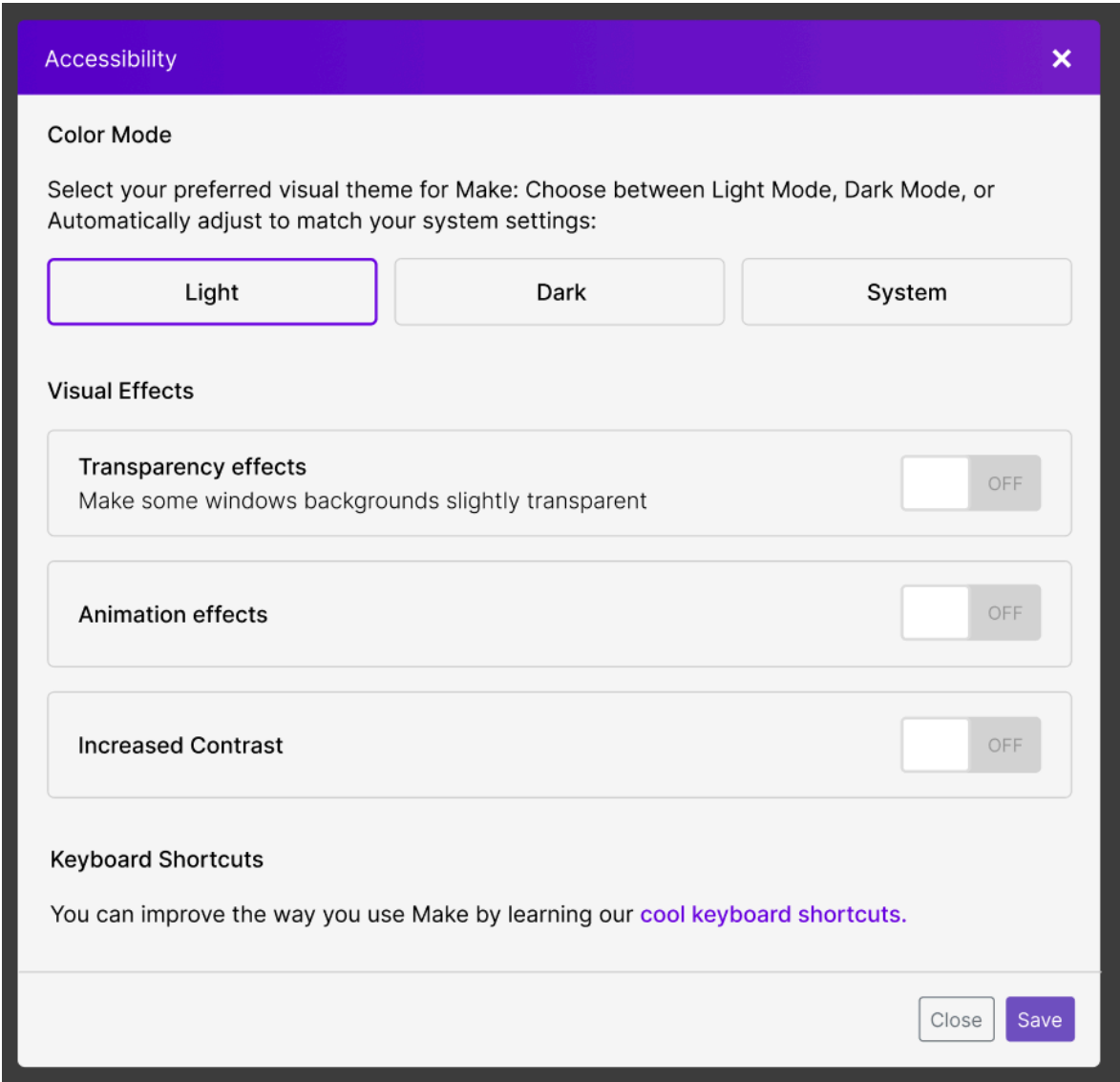
settings. These are purely illustrative at this stage, showing potential choices without active functionality.

In the Visual Effects Section, toggle placeholders within rectangles were featured where users can theoretically turn animation effects, transparency features, and interface contrast on or off. Like the color mode section, these toggles are non-functional and serve as visual representations of possible settings.

In the Keyboard Shortcuts Section, a text prompt was added, "You can improve the way you use Make by learning our cool keyboard shortcuts." This section indicates an interactive link to a subsequent window displaying all necessary keyboard shortcuts. However, at this point, the link is not active, and the list of shortcuts is not yet interactive but planned for future development.

Figure 12

Low-fidelity Customizable Settings Window prototype, 2024



At the top of this conceptual window, a static title bar labeled "Accessibility Settings" was placed, featuring a non-functional cross icon for closing the window. The bottom of the window hosts "Close" and "Save" buttons, which were also designed as non-interactive elements at this stage.

4.2.3. Keyboard Shortcuts Window

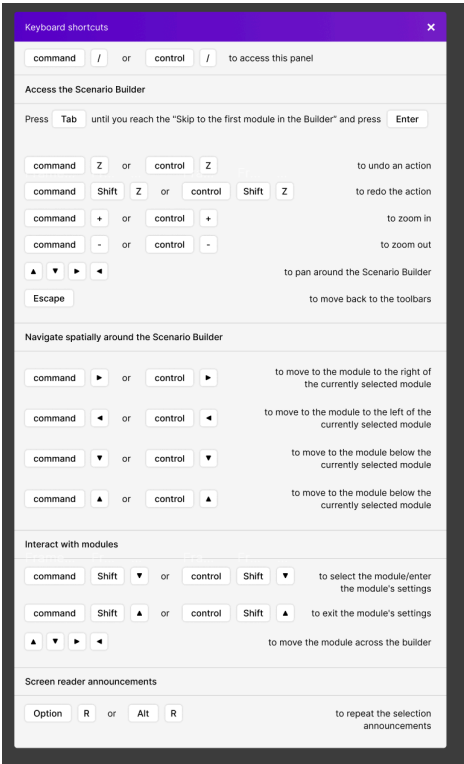
In designing the low-fidelity prototype for the last feature—the keyboard shortcut list—a comprehensive list of actions within the Make interface that currently lack keyboard accessibility was identified. These actions include accessing the shortcut window, entering and navigating within the Scenario Builder, zooming and panning, exiting back to the toolbars, selecting modules one by one, accessing and exiting module settings, moving modules, and repeating screen reader announcements.

Potential keyboard combinations for each action were then assigned, ensuring that every function could be executed through keyboard inputs alone, for both Apple and Windows users. This step involved theoretical mapping of shortcuts to actions without activating any interactive elements in the design process.

Following the planning phase, prototyping within Figma was initiated. A simple rectangle to serve as the layout for the shortcut list was drawn. Within this rectangle, the proposed keyboard key combinations alongside their descriptions were placed, organized by relevant functional groups to maintain clarity and ease of navigation. This was done in a purely visual format, focusing on the spatial arrangement and readability of the information.

Figure 13

Low-fidelity Keyboard Shortcuts Window prototype, 2024



At the top of the rectangle, a static title bar labeled "Keyboard Shortcuts" was added, including a non-functional cross icon, symbolizing a closing action. This icon, like the rest of the elements in the prototype, is designed for visual representation only and does not perform any actions at this stage.

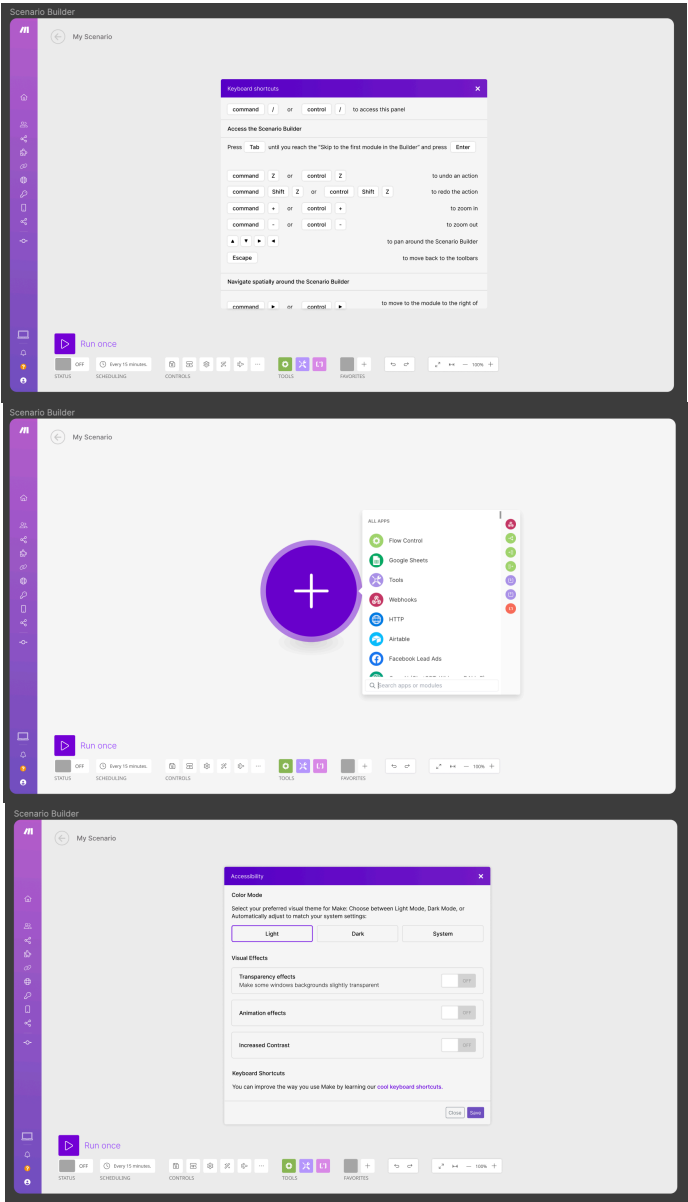
This methodical approach in Figma allows the necessary foundation for the keyboard shortcut feature in low fidelity to be laid out, emphasizing the arrangement and potential functionality of the shortcuts without delving into interactive elements just yet.

4.2.4.Preparing Prototypes for Testing

As the low-fidelity prototypes are prepared for testing, the focus is on setting up basic interactions that simulate how users would interact with the features on the screen.

Figure 14

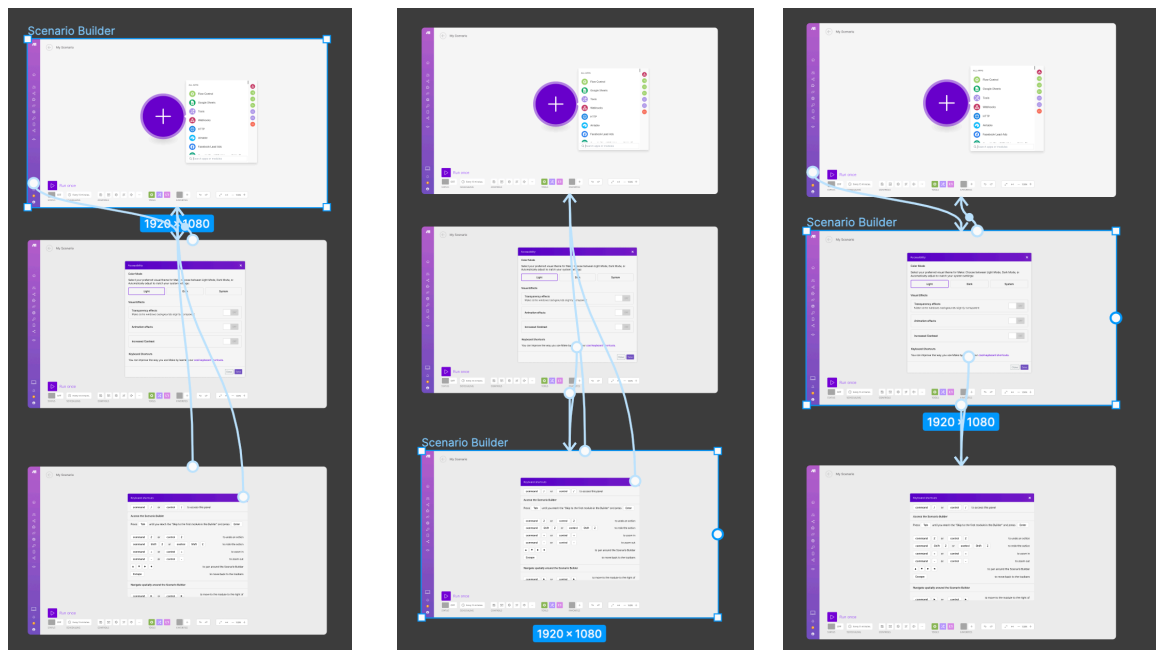
Low-fidelity prototypes of the Accessibility features, 2024



Although the prototypes are in low fidelity, adding these quick interactions is essential to gain valuable insights into user behavior and interface usability. In Figma, simple triggers and actions that mimic actual user interactions, such as clicking to access different settings and navigating through menus, will be configured. This setup will not only help in assessing the intuitiveness and efficiency of the navigation but also in understanding how users might engage with the accessibility features under real-world conditions.

Figure 15

Low-fidelity prototypes of the Accessibility features with interactions, 2024



By facilitating these preliminary interactions in the testing phase, the aim is to capture critical feedback early, enabling the refinement of the design before advancing to more complex interactive prototypes. This step ensures that the foundation of user interaction is solid and user-centered from the outset.

5. Testing and Reflection

In this chapter, we will explore the testing phase of my project, which, while not the primary focus of my thesis, plays a crucial role in validating the enhancements made to the Make interface. The goal of this testing was not to conduct exhaustive usability studies but to gather preliminary feedback on the low-fidelity prototypes that have been developed.

5.1. Testing of the low-fidelity prototypes

The testing procedure began with an introduction phase, where each session was initiated with a warm greeting and casual conversation to relax the participants. The role and purpose of the project were explained, setting a collaborative tone for the session.

During the next stage, information about each participant's use of similar platforms was gathered. By learning about their digital habits, the testing scenarios were tailored to be more relevant and engaging for each individual.

In the interactive testing phase, participants were provided access to the interactive prototype on a device. It was emphasized that the focus was on evaluating the application's functionality, not the user's abilities, which helped minimize any potential stress and encouraged open feedback. Participants were asked to complete specific tasks using the prototype while verbalizing their thoughts and actions. Observations and documentation of how they interacted with the prototype were carefully made, paying close attention to their use of the newly integrated accessibility features and noting any obstacles they encountered.

This structured yet flexible testing framework was integral in assessing how the accessibility improvements affected user interactions with Make, providing essential feedback for further refinement of the design.

5.2. Results of the Testing

In order to gain a well-rounded perspective on the new accessibility features in Make, it was decided that the prototype would be tested with a mix of users, some familiar with the Make platform and others encountering it for the first time. Out of five participants, three were not previously familiar with Make. This diversity allowed for an assessment of how intuitively new users could navigate the interface while also evaluating the improvement in usability for existing users. Interestingly, despite their varying levels of familiarity with Make, positive improvements in the accessibility features that were designed were noted by all participants. This feedback was encouraging as it suggested that the enhancements made the platform more user-friendly and accessible, significantly enhancing the user experience for both first-time and experienced users.

Valuable feedback was received from participants during the testing of the **Keyboard-Accessible Scenario Control** features. The proposed mini-map was well received and seen as a useful addition that could greatly improve navigation by providing a high-level view of the entire canvas. However, some users observed that the existing 'fit to screen' feature that aligns modules may overlap with the new functionality, suggesting a possible redundancy. Moreover, a few participants suggested relocating the zoom controls from the bottom to the top of the page to declutter the interface and enhance usability. These suggestions highlight the need to thoughtfully integrate new features while considering the existing layout and functionalities.

During testing of the **Customizable Settings Window** in the Make interface, valuable insights were shared by participants regarding its design and functionality. The placement of the settings icon received praise for its intuitiveness and ease of location. However, the majority of participants felt that the icon's immediate access was unnecessary and recommended it be nested within the Profile section for a more streamlined interface.

When discussing the customization of the scenario builder's color theme, users recommended integrating adjustment controls directly within the interface. This change would provide easier access and make the process more user-centric. Additionally, the options to toggle off animations and increase contrast were highly appreciated, as they improve readability and minimize distractions, enhancing the overall user experience.

However, some participants found it less intuitive to locate keyboard shortcuts through the settings window. To remedy this, it was suggested that if the shortcuts are primarily used within the scenario builder, a direct link to them should be prominently displayed within that interface. Lastly, the dark theme feature received universal acclaim for its usefulness, highlighting its importance in enhancing visual comfort for users across various environments.

During the testing of the **Keyboard Shortcuts Window**, participants acknowledged the utility of keyboard shortcuts but raised concerns about the clarity of presenting both Mac and Windows shortcuts on the same screen. They suggested that this could lead to confusion, especially for users unfamiliar with one of the operating systems. To enhance user experience and cater to different users' needs, it was recommended to create separate screens for Mac and Windows shortcuts. These screens would automatically adjust to match the user's operating system, ensuring a more intuitive and streamlined interaction.

Additionally, while participants appreciated the ability to move modules using arrow keys, they expressed a need for more precise control over these movements. Specifically, they desired the option to specify the exact number of pixels each movement should entail, allowing for finer positioning and alignment within the scenario builder. This feedback highlights the importance of detailed customization in accessibility features, enabling users to work with greater accuracy and efficiency.

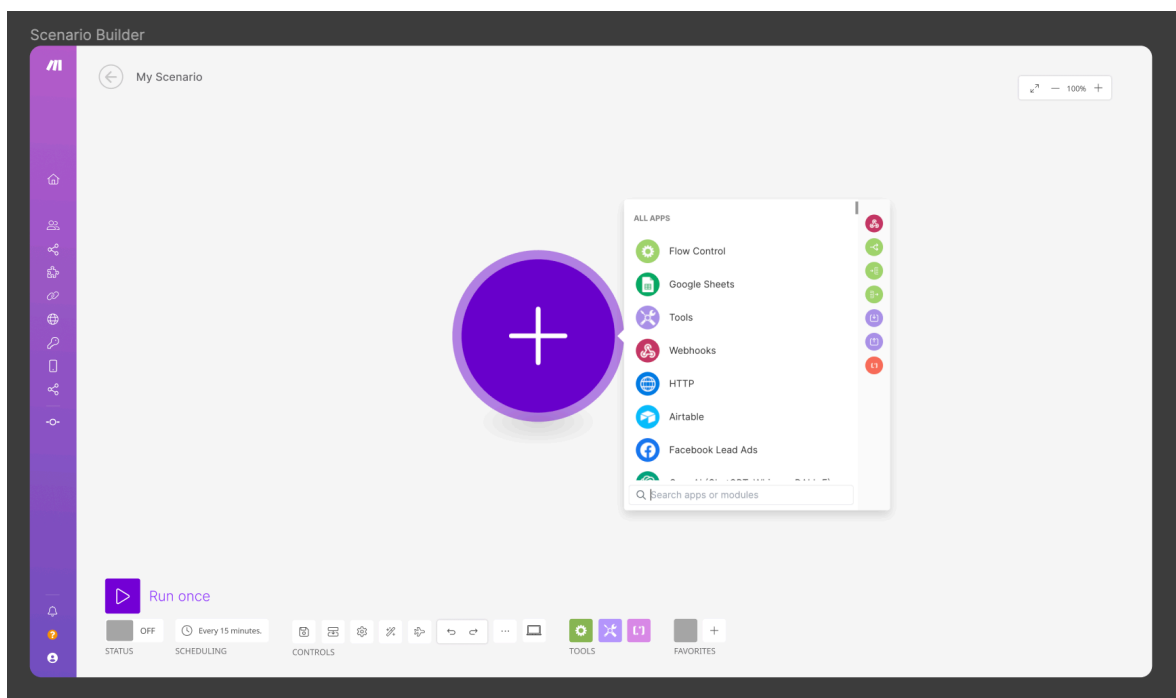
This feedback indicates that while the new features significantly enhance the accessibility and usability of the Make platform, there are opportunities for further refinement to meet user expectations more effectively, particularly in making controls more intuitive and contextually appropriate.

5.3. Implementing the Feedback into the design

Following the initial testing feedback, several changes were implemented in the Make interface that resonated as particularly sensible. These adjustments included repositioning the redo/undo buttons and altering the accessibility of the customizable settings window. By moving these elements to the controls section within the Scenario Builder, the aim was to streamline the user experience, making it easier for users to access keyboard shortcuts and adjust the application's appearance more efficiently. Additionally, the zoom controls were relocated to the top of the screen. This change was made to declutter the lower part of the interface, addressing user concerns about overcrowding and enhancing the overall usability of the workspace.

Figure 16

Changes in features based on the user feedback, 2024



Upon presenting these modifications to the same group of participants who tested the initial features, the response was immediately positive. The changes were expressed by the participants as making the interface more intuitive and user-friendly, validating the adjustments and confirming their effectiveness in improving the interaction experience. This feedback was encouraging and underscored the value of iterative design and user input in developing accessible and practical digital tools.

5.4. Testing Summary

To summarize, although testing was not the central objective of this thesis, the knowledge acquired from the sessions proved to be of great value. The input obtained from the participants validated the necessity and advantages of the features designed for the Make interface. Even though these features were not integrated into Make and were only conceived as part of the academic project, the favorable reaction to the prototypes is a crucial affirmation that the improvements are moving in the correct direction.

Conclusion

In conclusion, this bachelor thesis has addressed critical gaps in the accessibility features of Make, an online automation platform known for its user-friendly, no-code interface that supports complex workflows. While Make has made significant strides in integrating over 1500 applications, enhancing productivity and reducing the need for programming skills, it has fallen short in terms of accessibility, particularly for users who rely on assistive technologies.

The theoretical component of this thesis laid a solid foundation by examining the legislative, ethical, and business imperatives for web accessibility, alongside a comprehensive review of assistive technologies and accessibility analysis methodologies. This background helped to identify and understand the broader context of accessibility challenges faced by users of platforms like Make.

The practical segment of this thesis involved conducting an accessibility audit of Make, which pinpointed existing deficiencies and led to the development of three low-fidelity prototypes. These prototypes—enhanced navigation controls within the scenario builder, a customizable accessibility settings window, and a detailed list of keyboard shortcuts—were designed to directly address the needs highlighted in the audit. The design process was informed by theoretical insights and refined through user testing with five participants, whose feedback was instrumental in further refining these solutions.

The testing phase confirmed that the designed features are not only necessary but also highly valued by users, indicating a clear path toward improving Make's usability. The prototypes serve as a blueprint for future high-fidelity development and highlight the potential for engaging with users in an iterative design process.

This thesis has successfully achieved its primary goal of developing an accessible low-fidelity prototype of the Make platform. By integrating principles of accessible design and user-centered design, it has significantly strengthened digital inclusion. This work highlights the importance of inclusive design and user-centered development, emphasizing that accessibility should be integral to the development of digital tools. Moving forward, the insights and prototypes from this research provide a guide for future improvements, ensuring that Make becomes more accessible and inclusive for all users, thus broadening its reach and reinforcing its commitment to empowering users.

Bibliography

- Accessible contrast ratios and A-levels explained*. Stark Blog. (2021, August 10). <https://www.getstark.co/blog/accessible-contrast-ratios-and-a-levels-explained/>
- Boronines, A. (2012). *Color spaces for human beings*. Alexei Boronines Blog ATOM. <https://www.boronine.com/2012/03/26/Color-Spaces-for-Human-Beings/>
- Boschetti, A., & Massaron, L. (2018). *Python Data Science Essentials*. Packt Publishing.
- Companies that lead in Disability Inclusion Outperform Peers financially, reveals new research from Accenture*. Newsroom. (n.d.). <https://newsroom.accenture.com/news/2023/companies-that-lead-in-disability-inclusion-outperform-peers-financially-reveals-new-research-from-accenture>
- Conta, A. (2024). *The art and science of UX design: A step-by-step guide to designing Amazing User Experiences*. New Riders, an imprint of Peachpit Press (Pearson Education).
- Cunningham, K. (2012). *Accessibility handbook: Making 508 websites for everyone*. O'Reilly.
- Dam, R. F., & Siang, T. Y. (2024, May 4). *Stage 2 in the design thinking process: Define the problem and interpret the results*. The Interaction Design Foundation. <https://www.interaction-design.org/literature/article/stage-2-in-the-design-thinking-process-define-the-problem-and-interpret-the-results>
- Firth, A. (2019). *Practical web inclusion and accessibility: A comprehensive guide to access needs*. Apress.
- Georgakas, D. (2023). *A11Y unraveled: Become a web accessibility ninja*. Apress Media LLC.
- Gilbert, R. M., & Rateau, R. (2019). *Inclusive Design for a Digital World: Designing with accessibility in mind*. Apress.
- Horton, S., & Quesenbery, W. (2015). *A web for everyone: Designing accessible user experiences*. Rosenfeld Media.
- Matuzovic, M. (2024). *Web accessibility cookbook: Creating inclusive experiences*. O'REILLY MEDIA.
- Shariat, J., & Saucier, C. S. (2017). *Tragic design the impact of bad product design and how to fix it*. O'Reilly Media.
- The WebAIM million the 2024 report on the accessibility of the top 1,000,000 home pages*. WebAIM. (2024). <https://webaim.org/projects/million/>
- Tidwell, J., Brewer, C., & Valencia, A. (2020). *Designing interfaces: Patterns for effective interaction design*. O'Reilly Media.
- Webaim.org. (2024). *The WebAIM million the 2024 report on the accessibility of the top 1,000,000 home pages*. WebAIM. <https://webaim.org/projects/million/>

White, K., Lawton Henry, S., & Arch, A. (2024). *Easy checks – a first review of web accessibility*. Web Accessibility Initiative (WAI). <https://www.w3.org/WAI/test-evaluate/easy-checks/>

Appendix

Appendix A: Figma Prototypes of the Accessibility Features

<https://www.figma.com/file/oDiozapdjlxb7vH2AcmWMy/Ekaterina-Kirichuk-Thesis?type=design&node-id=0%3A1&mode=design&t=J66ygKRasFLHwINk-1>